The cut is *the product.*

Kuku editors watch every episode to find the 20–60 second moments that make a stranger stop scrolling. Promo Lab reads the timestamped screenplay, proposes candidate moments by **line ID**, resolves exact speech-safe timestamps in code, and evaluates every cut in a **blind cold-viewer pass** against naive baselines—ranking **52 verified promos across all 50 episodes** with explicit quality flags and failure diagnostics.

A PROMO MUST

- **Hook in 3 seconds** — a line or action that stops the scroll
- **Stand alone** — clear to someone who never saw the show
- **End unresolved** — cut while the question is still open
- **Point to exact time** — episode, start and end the editor can trust

01 WHY PROBLEM 1, AND WHAT I ASSUMED

I picked **Problem 1** because promo discovery is a weekly bottleneck across every show Kuku ships, and has a clear division of labor between deterministic code checks and LLM editorial judgment. Where the brief was silent, I assumed:

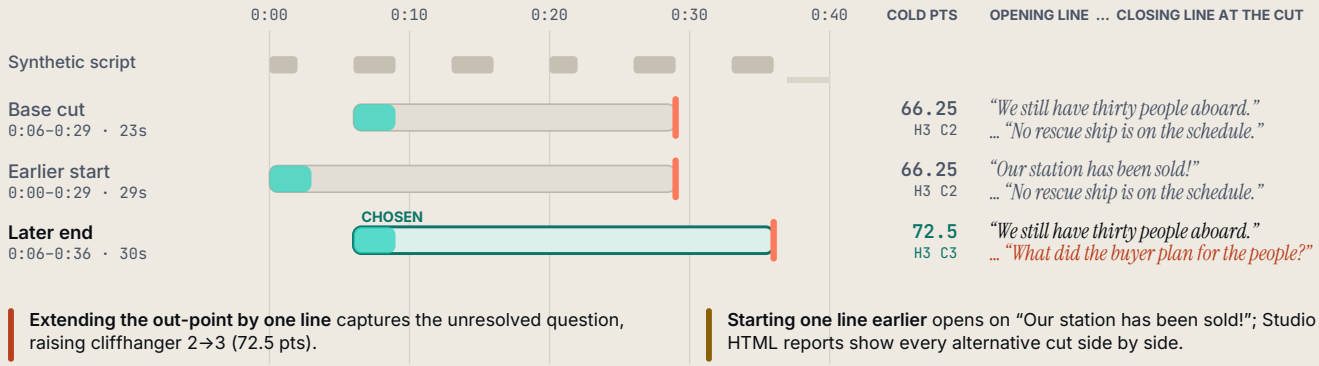
- **Script time proxies video time:** code never cuts a spoken line in half and flags timestamps for video frame check.
- **Standalone clarity != character biography:** a stranger only needs to grasp who is the underdog, who is the threat, and the immediate stake inside the cut.
- **Continuous first, multi-segment optional:** one continuous stretch is preferred, but joining up to 3 segments in story order is allowed when it sharpens a setup-to-payoff arc.
- **Deployability matters:** ranking penalizes high series-twist spoilers and caps each episode at 2 ranked cuts (keeping overlapping cuts as alternates).

02 THE INSIGHT: SPLIT PROPOSAL, RULES & COLD REVIEW

A single prompt that reads a full episode and outputs timestamps fails in two ways: the model **already knows the plot** (so it overrates clips that confuse a stranger), and it **hallucinates or drifts on timestamps** (in early runs, returning a 2-second clip or quoting lines outside its own cut).

Promo Lab separates the roles: (1) a **story-aware proposer** selects candidate cuts by **line IDs** (L0xx–L0yy); (2) **deterministic code** resolves exact script milliseconds, snaps edges so no overlapping speech is clipped, enforces 20–60 s duration, a 0–3 s hook, and verbatim quotes, and sends structured repair feedback when a rule breaks; (3) a **blind cold-viewer critic** reads only the re-timed clip (0.0 s = first frame), shuffled and unlabeled alongside naive episode-ending and random-window baselines; (4) **code** ranks and deduplicates.

03 WHY EXACT EDGES MATTER: SHIFTING A BOUNDARY BY ONE LINE CHANGES THE CLIFFHANGER



Extending the out-point by one line captures the unresolved question, raising cliffhanger 2→3 (72.5 pts).

Starting one line earlier opens on "Our station has been sold!"; Studio & HTML reports show every alternative cut side by side.

04 HOW IT WORKS: CODE ENFORCES STRUCTURE, MODELS JUDGE, EDITORS DECIDE

CODE	MODEL	CODE	MODEL	CODE	HUMAN
1. Parse	**2. Propose**	**3. Check & Repair**	**4. Cold Critic**	**5. Rank & Dedupe**	**6. Review Studio**
Strict line-accounting parser; numbers every timed speech, camera & action block.	Reads episode + story context; picks 1–3 cuts by line IDs with hook & cliffhanger.	Snaps to full speech; verifies 20–60s, 0–3s hook & exact quotes; 1 repair round.	Reads only 0.0s-retimed clips, shuffled & blind alongside 2 naive baselines/ep.	Weights H 30% · C 30% · S 25% · T 15% minus flags/spoilers; caps 2/ep.	promos.csv, HTML report & Studio web UI for final cut & video check.

Swappable model architecture: kuku promos speaks standard OpenAI /chat/completions JSON mode. Switch between **Gemini Flash** (config/promos.gemini.json), offline local **Qwen3-4B/8B via llama.cpp** (config/promos.local.json), or any interview API via KUKU_BASE_URL and KUKU_MODEL—zero code changes.

Ranked promos, blind benchmarks, and where it fails.

All **50 episodes** (3,727 timed blocks, 102.7 script minutes) were processed end-to-end. Every ranked promo passes deterministic boundary, duration (20–60 s), 3-second hook, and verbatim quote checks, and is scored cold in shuffled blind batches alongside two naive baselines per episode (last 40 s and random 40 s).

50/50
episodes analyzed · 57 valid cuts (10 repaired by code)

52 ranked
15 Tier-1 'post first' · 5 alternates · 8 rejected

62.0 vs 49.1
blind cold score: tool best/ep vs ending (49.1) & random (46.4)

31/43 wins
tool best/ep beats naive ending (7 ties, 5 losses) in blind test

05 TOP 8 RANKED PROMOS TO POST FIRST (CHARACTER NAMES GENERALIZED HERE FOR PUBLIC SAFETY; EXACT QUOTES IN PROMOS.CSV)

RANK · ID	EPISODE & TIMECODES	COLD SCORE	OPENING HOOK (FIRST 3 SECONDS)	UNRESOLVED CLIFFHANGER AT CUT
#1 E32-e67058	E32 119.5–154.4s 34.9 s · continuous	100 H5 C5 S5 T5	The noble rival snaps entirely, screaming at her prisoner while brandishing a red-hot iron, immediately...	The branding iron is held mere millimeters from the heroine's eye; will she be permanently disfigured or killed?
#2 E41-2545d3	E41 2–13s+81–96s 26.3 s · multi-seg	100 H5 C5 S5 T5	The noble rival halts her own wedding to publicly humiliate the protagonist, creating immediate tension.	The heroine sees her savior emerge from the smoke, leaving the viewer wondering if she will be rescued.
#3 E39-7110f5	E39 50–67s+86–98s 29.8 s · multi-seg	92.5 H4 C5 S5 T5	The scarred protector stares at his scarred, unkempt reflection, immediately signaling a major transformation is...	The scarred protector vows to bring his wife home, leaving the viewer anticipating the confrontation at the...
#4 E38-3067c8	E38 64.5–87.9s 23.4 s · continuous	88.8 H4 C5 S5 T4	The heroine's eyes snap open in terror as the wedding bells toll, signaling her impending death.	The scene cuts to black on a high-tension sting, leaving the viewer in suspense as the wedding begins.
#5 E50-2e5702	E50 2–16s+70–80s 23.5 s · multi-seg	82.8 H4 C5 S5 T4	The noble rival and the priest are shown in bed, immediately establishing a scandalous, forbidden affair that...	The witness witnesses the secret affair, leaving the viewer to wonder if she will expose them or be silenced.
#6 E23-cd0b35	E23 79.0–131.5s 52.5 s · continuous	82.5 H4 C5 S4 T4	The heroine bluntly accuses someone of poisoning the noblewoman, immediately raising the stakes of the mystery.	The door is opening while the heroine is caught red-handed with the antidote, creating a perfect 'trapped'...
#7 E01-cdc630	E01 11.0–63.0s 52.0 s · continuous	81.2 H4 C4 S5 T4	A cold matriarch stops the bride halfway down the aisle with an ominous warning.	The groom calmly breaks off the marriage with zero empathy right at the altar.
#8 E03-858302	E03 58.3–96.5s 38.2 s · continuous	75 H5 C3 S4 T4	A matriarch shouts for guards to drag out a bride, triggering an immediate physical scuffle.	After KOing a guard, the menacing stranger warns the rest of the chapel not to touch the bride again.

06 WHERE IT PICKS BADLY (HONEST FAILURE MODES)

1. Complete-beat endings vs true cliffhangers. In **8/57** valid candidates, the proposer cuts right after a sharp retort finishes rather than before the reaction. The blind cold critic catches this ("feels like a complete scene beat rather than an open hook") and scores cliffhanger ≤2/5.

2. Multi-segment jump cuts. When the proposer joins two moments (**16** cuts, e.g., E02-0149e3 jumping from 0:12 to 1:09), the cold critic flags the abrupt mid-clip transition as disorienting without a visual transition.

3. Late-series plot load & spoilers. In later episodes, candidates can reveal major arc twists or rely on nobility/ledger backstory; code penalizes high spoiler risk (-6 pts) so early-episode hooks rank first.

4. Earlier V1–V7 local-model failure modes. On Qwen3-4B/8B, free-form generation returned a **2-second** cut twice and quoted dialogue outside its own cut in **15/50** proposals (V7 audit), while V3's strict lexical gate treated a kneeling proposal question as "answered" (0/50 accepted).

07 QUALITY CHECKING: BLIND BASELINES & ITERATION HISTORY

Shuffled blind baseline test. Every episode's candidates were shuffled unlabeled with its naive last-40s cut and a random 40s window. **Tool best/ep** averages **62.0** (H 3.14, C 3.58, S 3.93, T 3.21) vs **60.0** across all tool cuts, **49.1** for episode endings, and **46.4** for random windows.

Why proposer self-scores aren't enough. Proposer self-score vs blind cold critic correlates only weakly (Spearman **r = 0.293**): a proposer that knows the full episode overrates cuts that need outside context.

Progression across versions. **V1** (Qwen3-8B, T4): 10 unverified picks · **V3/V6/V7**: strict lexical & quote-location audits (found 15/50 out-of-cut quotes, 0/50 accepted) · **V8**: line-ID boundary resolution + repair loop + blind cold critic → 52 ranked verified cuts.

Automated test suite & leak scan. **139 unit tests** cover parser, zip safety, quote grounding, Studio UI, and V8 boundary/ranking invariants; a 24,554-shingle scanner verifies zero screenplay text in public assets.

08 SCALE AND COST

Measured (V8, 50 episodes): 53 total model requests (0.35M tokens) via multi-episode batching and worker concurrency—**$0.00** on free-tier Gemini Flash (~$0.03 at paid Flash rates) and ~4.4 min wall-clock. **Offline fallback:** local Qwen3-8B on a free Kaggle T4 took 47.7 min (0.57M tokens, $0). At 100 shows/week, batched Flash costs <$4/week.

09 AI TOOLS & WHERE AI GOT IT WRONG

Tools used: Gemini, Claude, and Codex for code, audits, Studio UI, and PDF design; **Gemini Flash** (V8) and **Qwen3-4B/8B** (V1–V6) as in-pipeline models.
Where AI got it wrong: (1) Local model returned a 2-second clip twice and quoted out-of-cut lines (15/50)—fixed by line-ID selection + code boundary check; (2) V3 regex gate marked a marriage proposal question as "answered"—fixed by blind cold-viewer critique.

10 HOW WE'D KNOW IT WORKED IN PROD

1. **Editor time saved:** minutes to lock 5 promos/show vs watching all 50 episodes cold.
2. **Boundary precision:** % of Tier-1 picks exported to NLE without >2 s trim adjustments.
3. **Video timing audit:** measure script-timestamp vs rendered audio/video drift.
4. **Live ad lift:** 3-second thumb-stop rate and tap-to-watch CTR on Reels/TikTok/Shorts.

11 CASE STUDIES FROM THE 50-EPISODE RUN: ONE AUTO-REPAIRED CUT, ONE REJECTED CUT, ONE BLIND-CRITIC DEMOTION

Auto-repaired by code -> Tier-1 (#11)

E26-9958a9 · Ep 26 · 34.4 s · 75.0 pts

Initial proposal: boundary violation flagged by code. **After 1 repair round:** adjusted to L034–L051 (34.4 s), opening on a close-up visual reveal of a noble signet ring on a maid's hand and cutting as the noblewoman clutches the ring in shock. **Blind critic:** H3 · C5 · S4 · T4.

Caught & rejected at 60.6 s

E46-8ed1bb · Ep 46 · 60.6 s · Rejected

Initial proposal: 2-segment cut totaling 71.8 s (>60 s cap). **After 1 repair round:** model trimmed Segment 1 to L008–L026, bringing total duration to 60.6 s—still 0.6 s over the 60 s hard rule. Logged to rejected.json rather than silently passing.

High self-score demoted by cold critic

E17-91e4b3 · Ep 17 · 41.0 s · 23.8 pts

Proposer self-score: rated high because the proposer knew the prior attack backstory. **Blind cold critic:** scored 23.8/100 and flagged "Who are the men he is protecting her from, and why is she bandaged?"—demoted to weak.